

Grade Eight Science - Magnetism and Static Electricity Activities

Activity 1:

Rub an ebonite rod with fur. Touch the rod to the ball of the electroscope. **Outline** the behaviour of the electroscope leaf below

Outline of the behaviour: The electroscope leaf will be extended outward at the end of the activity.
Scientific Explanation of what is happening: Fur is below ebonite on the Triboelectric series. Therefore, fur has a lower electron affinity (EA) than ebonite. This means that it is easier to remove electrons from fur than it is from ebonite. Both fur and ebonite start out as neutral (equal number of positive and negative charges). When fur is rubbed against an ebonite rod (charging by friction) the fur will give electrons to the rod making the fur positive and the rod negative. When the negative ebonite rod comes in contact with the neutral electroscope, the rod gives negatives to the electroscope to try and make itself neutral. Removing the rod leaves a neutral charge on the rod and a negative charge on the electroscope.

Activity 2:

Rub a glass rod with wool. Put the rod close to the paper confetti. **Outline** the behaviour of the paper confetti below

Outline of the behaviour: The paper confetti will move toward the rod. Some of the confetti may touch the rod and fall back down
Scientific Explanation of what is happening: The glass rod and wool start out neutral. Since glass has a lower EA (lower on the Triboelectric Series) than wool, rubbing wool against a glass rod (charging by friction) will move electrons from the glass onto the wool making the wool negative and the glass positive. The paper confetti is neutral. When a positive rod comes close to the paper confetti it causes the charges on the confetti to polarize, meaning the charges will separate with the negatives on the confetti moving closer to the positive rod (attraction), and the positives on the confetti moving away from the rod (repulsion) Since confetti is light, the force of attraction is strong so the forces can lift the confetti toward the rod. If the paper confetti (neutral) hits the positive rod, the rod can take negatives from the confetti to neutralize itself leaving the confetti positive.

Activity 3:

Rub a plastic rod with styrofoam. Put the rod near the ball of a neutral electroscope.

Outline the behavior of the electroscope leaf below

Outline of the behaviour: When the rod is placed near the electroscope, the leaf of the electroscope will extend outward. When the rod is taken away from the electroscope the leaf will go back down.
Scientific Explanation of what is happening: Plastic is below styrofoam on the Triboelectric series. This means that if the plastic rod (low EA) is rubbed with styrofoam (high EA) the plastic rod will give electrons to the styrofoam. This makes the plastic rod positive and the styrofoam negative. When the positive plastic rod is placed near the ball of the neutral electroscope, the force of attraction will cause the charges on the electroscope to polarize. The negative charges that are attracted to the positive rod will go to the ball of the electroscope, while the positive charges move down to the leaf (like charges repel). The leaf extending outwards is the repulsion. The negatives are moving as far away from the positive rod as they can get because like charges repel.

Activity 4:

Use the magnet provided to see which of the following materials are magnetic and

outline your answers below

- A. Iron
- B. Aluminum
- C. Paper clip

A. Iron	B. Aluminum	C. Paper Clip
Observation: Magnetic - attracted to the magnet	Observation: Not magnetic - not attracted to the magnet (no movement)	Observation: Magnetic - attracted to the magnet
Scientific Explanation: A magnetic material is made up of tiny areas of spinning electrons called domains. Iron is magnetic because the electrons are aligned with the magnetic field lines rather than randomly arranged. When the electrons are aligned the area is magnetic. When the electrons are randomly arranged the magnetic fields cancel each other out (not magnetic)	Scientific Explanation: Aluminum is not magnetic because the electrons in the domains are randomly arranged, rather than aligned. When the electrons are randomly arranged, rather than aligned the magnetic fields cancel each other out (not magnetic)	Scientific Explanation: A magnetic material is made up of tiny areas of spinning electrons called domains. A paper clip is magnetic because the electrons are aligned with the magnetic field lines rather than randomly arranged. When the electrons are aligned the area is magnetic. When the electrons are randomly arranged the magnetic fields cancel each other out (not magnetic)

Activity 5:

Using the bar magnets provided, complete each of the following actions and **outline** the behavior of the magnets below

- A. Put the North pole of each magnet near one another
- B. Put the South pole of each magnet near one another
- C. Put the North pole of one magnet near the South pole of another magnet

A: North - North	B. South - South	C. North - South
Observation: The magnets move away from each other (repulsion) Scientific Explanation: Attraction and repulsion is caused by the behaviour of the magnetic field lines. When magnets repel, like poles cause the field lines to clash and compress. When field lines clash the like poles are pushed away from one another	Observation: The magnets move away from each other (repulsion) Scientific Explanation: Attraction and repulsion is caused by the behaviour of the magnetic field lines. When magnets repel, like poles cause the field lines to clash and compress. When field lines clash the like poles are pushed away from one another	Observation: The magnets move toward one another (attraction) Scientific Explanation: Attraction and repulsion is caused by the behaviour of the magnetic field lines. When magnets attract, unlike poles, allow field lines to connect and flow smoothly allowing the poles to come together rather than be pushed apart

Activity 6:

Move the bar magnet around the iron filings box. **Outline** the behaviour of the iron filings

Outline of the behaviour: The iron filings move toward the magnet. The iron filings will follow the magnet as you move the magnet around the box of iron filings
Scientific Explanation of what is happening: When a bar magnet is placed near a box of iron filings the iron filings become induced magnets becoming magnetized by the presence of the magnetic field. Because they are needle shaped (just like the needle on a compass) they align end to end with the magnetic field lines. You can't see the magnetic field lines, but you know they are there because of the behaviour of the iron filings